# Robed Beauvile-Owen, Ph.D.

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## EXPERIENCE

**Founder & Data Scientist** — *AniPoint Site Research* Pittsburgh, PA | May 2026 - Present

- Founded a location intelligence consultancy for retail and service businesses and identified $1.5M+ expansion opportunities.
- Designed a revenue driver tree (traffic × capture rate × ticket price) and a margin model that turned client profit goals into break-even targets.
- Estimated the incremental revenue of new sites with observational causal inference (Double ML, causal forests) on a 12-year county panel, with robustness checks (difference-in-differences) and out-of-sample back-tests against real sites.
- Developed scenario planning (low/base/high) and Monte Carlo risk models (probability of loss, VaR, CVaR) on reproducible, tested Python workflows (150+ unit tests) that cut turnaround by 91%.

**Senior Data Scientist** — *Limetree* Dallas, TX (Remote) | September 2024 - April 2026

- Owned the consumer funnel KPI framework and driver tree linking gross lead rate (GLR) to member rate and cost per member with strategy, finance, and engineering, using a causal graph (DAG), tested on historical data, to identify GLR as the lever to model and test.
- Translated financial goals into team targets with a funnel sensitivity model showing the cost-per-member target required GLR of at least 14.2%.
- Redefined GLR as a certified metric for a new client missing key lead markers, working with data engineering to date leads by last response, credit them to campaigns, and flag broker influence.
- Developed uplift and segmentation models on millions of HIPAA-protected records, used in 87% of campaigns, cutting ad spend by $10K per campaign and moving 12% of one client's budget to its top GLR segment.
- Created an in-house experimentation and reporting platform that data science and strategy used to validate lift and to run monthly reporting, weekly analysis, Power BI ingestion, and metric monitoring.
- Used the finding that claims and ticket complaints drove churn to improve the churn model, and re-architected its platform on Spark and Databricks, cutting runtime by 48% and memory by 23%.

**Founding Machine Learning Engineer** — *eParts Services LLC* Pittsburgh, PA | September 2023 - September 2024

- As the first ML hire, defined success metrics and drift monitoring for search and pricing models, and mentored the team on ML practices.
- Shipped a regression pricing API (C#, Azure, 10,000+ daily requests) that cut false and anomalous shipping prices by 13%, reducing cost leakage and holding a stable 18% contribution margin.
- Cut search dissatisfaction by 12% and raised platform engagement by 16% with custom Elasticsearch indexing and TF-IDF weighting.

**Data Scientist** — *Limetree* Dallas, TX (Remote) | April 2022 - September 2023

- Improved the XGBoost GLR model (AUC-ROC up 3%) and validated it with an A/B test. SHAP surfaced the drivers (income, age, children in the household) and uplift models isolated the creative's effect.
- Validated GLR out of sample as a leading indicator of member rate 60 days ahead ($R^2 = 0.28$), and improved gross lead rates by 18%.
- Led monthly monitoring of up to 10 production models across clients (accuracy, drift, forecast vs. actual) for leadership, and built a company-wide self-service dashboard on curated Snowflake and Databricks tables.

**Graduate Researcher & Teaching Assistant** — *University of Pittsburgh* Pittsburgh, PA | August 2017 - May 2022

- Produced 3 original results in optimization and operator theory, and taught statistics to classes of 75-200 students.

## TECHNICAL PROJECTS

**Customer LTV & Churn Survival Modeling** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Simulated 1M customer lifetimes with tenure-dependent churn (10%, 5%, then 2% monthly) to show the LTV = ARPU / churn identity holds only under constant churn.
- Estimated expected lifetime and LTV from the empirical survival curve ($E[T] = \int S(t)dt$), matching the simulated mean.

**Mobile Game Retention A/B Test** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Analyzed a test moving a mobile game's first progress gate from level 30 to 40, measuring 1- and 7-day retention and rounds played.
- Set Type I/II error rates, sized samples, removed outliers with IQR, and weighed statistical against practical significance.

**Medicare Claims Intelligence & Fraud Detection** github.com/beauvilerobed/data-science-machine-learning-lab-and-compendium

- Built a dbt warehouse (4 staging models, 5 marts, 26 data tests) over 790K CMS claims, with tests that caught 11,253 claims missing a service date.
- Beat a seasonal-naive forecast in a 35-origin rolling backtest (MASE 0.62 vs. 1.06), reached 81% precision on planted fraud at review capacity, and built an out-of-time XGBoost risk model (SHAP, MLflow) finding 46% more high-cost members.

**Starbucks Rewards: Predicting Offer Responses** github.com/beauvilerobed/predict-consumer-responses

- Predicted how rewards members respond to BOGO, discount, and informational offers from transaction and demographic data, using XGBoost on AWS SageMaker.

## EDUCATION

**Ph.D. in Mathematics** — *University of Pittsburgh* 2017 - Dec 2023

**Dissertation:** A Minimal Invariant Set, Frames, and Operators

**B.S. in Mathematics** — *Florida State University* 2014 - 2017

**Minor:** Philosophy | **GPA:** 3.90

## TECHNICAL SKILLS

**Causal Inference & Experimentation:** A/B Testing, Experimentation Platforms, CUPED, Power Analysis, Observational Causal Methods, Double ML, Causal Forests, Difference-in-Differences, Uplift Modeling, Causal Graphs (DAGs), Robustness Checks

**Modeling & Statistics:** Regression, Classification, XGBoost, Time Series Forecasting, Survival Analysis, Monte Carlo Simulation, Graph Models, Multi-Metric Modeling, SHAP, Model Monitoring & Drift

**Data & Tools:** SQL, Python (pandas, statsmodels, scikit-learn, econml), R (survey analysis), Snowflake, Databricks, Spark, dbt, Curated Tables & Data Marts, Self-Service Analytics, Reproducible & Tested Workflows, AWS, Azure, Power BI, Tableau, Git

**Collaboration:** Cross-Functional Partnership (Product, Finance, Engineering, Strategy), Stakeholder Communication, Method Documentation, Mentoring

**Metrics & Planning:** KPI Frameworks, KPI / Driver Trees, Certified Metric Definitions, Leading Indicators, Consumer Product Analytics (Acquisition, Activation, Engagement, Retention, Monetization), CAC, LTV / ARPU, Contribution Margin, Scenario Analysis, Target Setting, Budget Allocation, Forecasting